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(hadoop@kali)-[~]  
$ mkdir exp2
```

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(hadoop@kali)-[~]  
$ cd exp2
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(hadoop@kali)-[~/exp2]  
$ ls  
dataset.txt
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(hadoop@kali)-[~/exp2]  
$ nano mapper.py
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(hadoop@kali)-[~/exp2]  
$ nano reducer.py
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(hadoop@kali)-[~/exp2]  
$ start-all.sh  
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.  
WARNING: This is not a recommended production deployment configuration.  
WARNING: Use CTRL-C to abort.  
Starting namenodes on [localhost]  
Starting datanodes  
Starting secondary namenodes [kali]  
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true  
Starting resourcemanager  
Starting nodemanagers
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```
(hadoop@kali)-[~/exp2]  
$ hdfs dfs -mkdir /exp2  
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true  
  
(hadoop@kali)-[~/exp2]  
$ hdfs dfs -put dataset.txt /exp2  
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
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(hadoop@kali)-[~/exp2]
$ hadoop jar $HADOOP_STREAMING -input /exp2/dataset.txt -output /exp2/output -mapper ~/exp2/mapper.py -reducer ~/exp2/reducer.py
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
packageJobJar: [/tmp/hadoop-unjar5095794439998385058/] [] /tmp/streamjob9814844253380862376.jar tmpDir=null
2024-08-31 05:08:08,890 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-08-31 05:08:09,197 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-08-31 05:08:09,908 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1725094826814_0003
2024-08-31 05:08:10,468 INFO mapred.FileInputFormat: Total input files to process : 1
2024-08-31 05:08:10,661 INFO mapreduce.JobSubmitter: number of splits:2
2024-08-31 05:08:11,120 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1725094826814_0003
2024-08-31 05:08:11,120 INFO mapreduce.JobSubmitter: Executing with tokens: []
2024-08-31 05:08:11,516 INFO conf.Configuration: resource-types.xml not found
2024-08-31 05:08:11,517 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2024-08-31 05:08:11,641 INFO impl.YarnClientImpl: Submitted application application_1725094826814_0003
2024-08-31 05:08:11,711 INFO mapreduce.Job: The url to track the job: http://kali:8088/proxy/application_1725094826814_0003/
2024-08-31 05:08:11,713 INFO mapreduce.Job: Running job: job_1725094826814_0003
2024-08-31 05:08:26,281 INFO mapreduce.Job: Job job_1725094826814_0003 running in uber mode : false
2024-08-31 05:08:26,299 INFO mapreduce.Job: map 0% reduce 0%
2024-08-31 05:08:42,128 INFO mapreduce.Job: map 100% reduce 0%
2024-08-31 05:08:55,399 INFO mapreduce.Job: map 100% reduce 100%
2024-08-31 05:08:57,510 INFO mapreduce.Job: Job job_1725094826814_0003 completed successfully
2024-08-31 05:08:57,810 INFO mapreduce.Job: Counters: 55
File System Counters

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FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=83481
HDFS: Number of bytes written=96
HDFS: Number of read operations=11
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0

Job Counters
Killed map tasks=1
Launched map tasks=2
Launched reduce tasks=1
Data-local map tasks=2
Total time spent by all maps in occupied slots (ms)=27207
Total time spent by all reduces in occupied slots (ms)=10965
Total time spent by all map tasks (ms)=27207
Total time spent by all reduce tasks (ms)=10965
Total vcore-milliseconds taken by all map tasks=27207
Total vcore-milliseconds taken by all reduce tasks=10965
Total megabyte-milliseconds taken by all map tasks=27859968
Total megabyte-milliseconds taken by all reduce tasks=11228160

Map-Reduce Framework
Map input records=365
Map output records=10220

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Input split bytes=180
Combine input records=0
Combine output records=0
Reduce input groups=12
Reduce shuffle bytes=102100
Reduce input records=10220
Reduce output records=12
Spilled Records=20440
Shuffled Maps =2
Failed Shuffles=0
Merged Map outputs=2
GC time elapsed (ms)=272
CPU time spent (ms)=5510
Physical memory (bytes) snapshot=666062848
Virtual memory (bytes) snapshot=8097644544
Total committed heap usage (bytes)=384827392
Peak Map Physical memory (bytes)=261619712
Peak Map Virtual memory (bytes)=2695987200
Peak Reduce Physical memory (bytes)=164155392
Peak Reduce Virtual memory (bytes)=2706378752

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
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WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=83301
File Output Format Counters
  Bytes Written=96
2024-08-31 05:08:57,821 INFO streaming.StreamJob: Output directory: /exp2/output

(hadoop@kali)-[~/exp2]
$ hdfs dfs -cat /exp2/output/part-00000
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
01      26.5
02      26.6
03      29.1
04      30.8
05      31.1
06      33.6
07      38.5
08      40.2
09      36.5
10      36.9
11      27.6
12      25.9
```